

S1 appendix for

The Impact of Groupwork on Undergraduate State Anxiety in Science: *“It’s like willingly going into a dark forest where anything bad can happen.”*

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Aspects hypothesized to alleviate undergraduate state anxiety and the literature supporting each.

Aspect	Supporting literature
Having other students support you academically	(Cooper et al., 2018; Downing et al., 2020).4/20/26 3:32:00 PM
Realizing that other students struggle with content too	(Cooper et al., 2018).
Being able to work with other students who you know	(Baldock & White, 2026; Cooper et al., 2018; Downing et al., 2020; Strauss et al., 2011).
Having another student validate your contribution to a discussion	(Downing et al., 2020).
Forming new personal connections with other students	(Baldock & White, 2026; Strauss et al., 2011; Xethakis et al., 2024).

Aspects hypothesized to exacerbate undergraduate state anxiety and the literature supporting each.

Aspect	Supporting literature
Being the center of attention	(Cooper et al., 2018; Downing et al., 2020; England et al., 2019).
Realizing that you may struggle with content more than other students	(Cooper et al., 2018).
Having your grade depend on other students' contributions	(Burdett, 2003; Chang & Brickman, 2018; Chi & Kadandale, 2022).
Working with other students who you do not know	(Baldock & White, 2026; Cooper et al., 2018; Downing et al., 2020; Strauss et al., 2011; Xethakis et al., 2024).
Making a mistake in front of other students	(Downing et al., 2020; Luan et al., 2022; Xethakis et al., 2024).
Feeling like you have talked too much	(Cooper et al., 2018).
Feeling like you have talked too little	(Cooper et al., 2018).
A student responding rudely to your contribution	(Cooper et al., 2018; Downing et al., 2020; Luan et al., 2022)
Other students treating you differently than they would treat others	(Baldock & White, 2026; Downing et al., 2020; Luan et al., 2022; Xethakis et al., 2024).
Being asked to initiate social interactions	(Xethakis et al., 2024).
Not being able to be your authentic self	(Van den Bosch & Taris, 2018; Van Den Bosch & Taris, 2014)

Learning domains hypothesized to be impacted by undergraduate state anxiety and the literature supporting each.

Learning domains	Supporting literature
Remember content that you learned	(Dowker et al., 2016; Hashempour & Mehrad, 2014; Lau et al., 2022).
Stay focused	(Baldock & White, 2026; Cooper et al., 2018).
Be motivated to engage in discussions	(Downing et al., 2020).
Be patient with others	(Rozi, 2021).
Communicate your thoughts	(Cooper et al., 2018; Downing et al., 2020; Xethakis et al., 2024).
Be willing to contribute	(Cooper et al., 2018; Downing et al., 2020).
Understand what others are saying	(Xethakis et al., 2024).
Solve problems	(Cooper et al., 2018).4/20/26 3:32:00 PM
Feel confident about your intelligence	(Downing et al., 2020; Luan et al., 2022).

Survey

Please think of one college science course that you are taking that involves group work. For the remainder of the survey, all questions will be about your experience in that specific class.

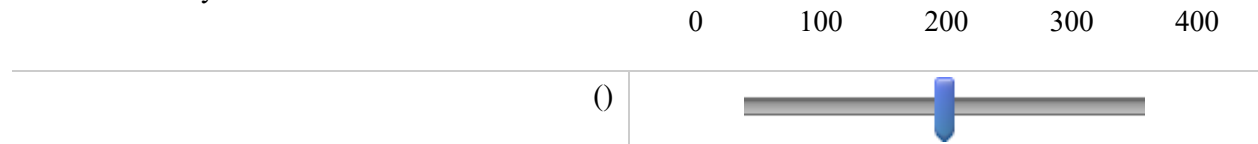
What is the primary subject of the course?

- Astronomy
- Biology
- Biochemistry
- Chemistry
- Environmental science
- Geosciences
- Physics
- Other _____

What level is this course?

- Upper level (300/400)
- Lower level (100/200)

About how many students are in this course?



What is the average number of other students (not counting yourself) who you work with during groupwork in this class?

- (1)
- (2)
- (3)
- (4)
- (5) or more

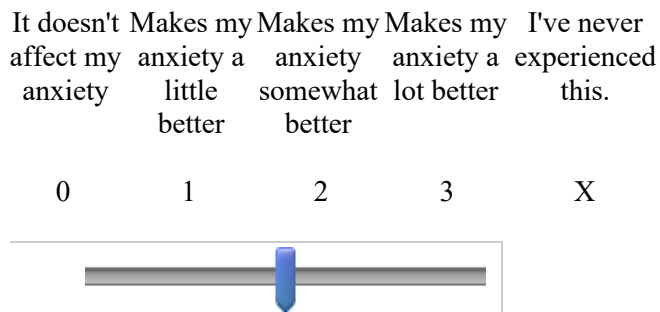
Everyone has feelings of anxiety to some extent. For some students, these feelings are short-lived and do not affect them much in class. For others, they are more severe.

Please fill out the following questions about your anxiety during the past two weeks in your college science course. Over the last 2 weeks, how often have you been bothered by the following problems in your college science course?

	Not at all (0)	Several days (1)	Over half the days (2)	Nearly every day (3)
Feeling nervous, anxious, or on edge	☐	☐	☐	☐
Not being able to stop or control worrying	☐	☐	☐	☐
Worrying too much about different things	☐	☐	☐	☐
Trouble relaxing	☐	☐	☐	☐
Being so restless that it's hard to sit still	☐	☐	☐	☐
Becoming easily annoyed or irritable	☐	☐	☐	☐
Feeling afraid as if something awful might happen	☐	☐	☐	☐

The next set of questions will ask you to what extent aspects of groupwork make your feelings of anxiety **better**. When we refer to feelings of anxiety, we are talking about feeling nervous, anxious, on edge, afraid, or worried. Please answer the following questions based on your experience in the one science course you indicated at the beginning of the survey.

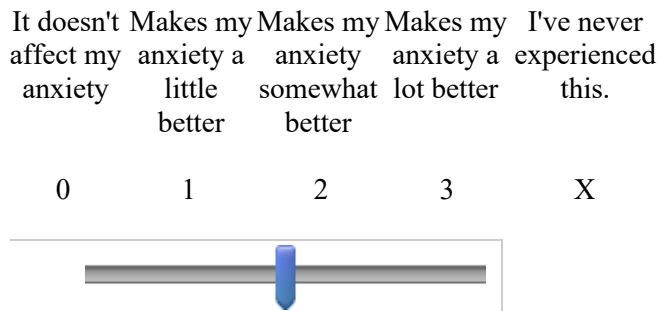
During groupwork, how, if at all, **does having other students support you academically** (e.g., answer your questions, discuss topics with you, help you understand content) affect your feelings of anxiety?



Students who answered 2 or greater were displayed this question:

Please explain why **having other students support you academically** makes your anxiety better.

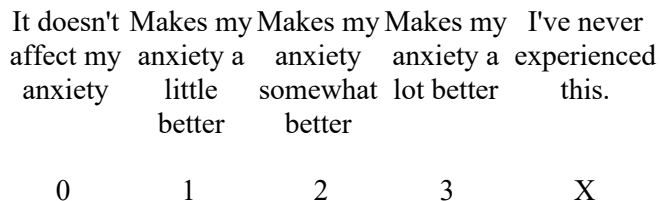
During groupwork, how, if at all, **does realizing that other students struggle with content too** affect your feelings of anxiety?



Students who answered 2 or greater were displayed this question:

Please explain why **realizing that other students struggle with content too** makes your anxiety better.

During groupwork, how, if at all, **does being able to work with other students who you know** affect your feelings of anxiety?





Students who answered 2 or greater were displayed this question:

Please explain why **being able to work with other students who you know** makes your anxiety better.

During groupwork, how, if at all, **does having another student validate your contribution to a discussion** (e.g., say ‘that’s a good point/question, that’s interesting, I agree with you, etc...?’) affect your feelings of anxiety?

It doesn't affect my anxiety	Makes my anxiety a little better	Makes my anxiety somewhat better	Makes my anxiety a lot better	I've never experienced this.
0	1	2	3	X



Students who answered 2 or greater were displayed this question:

Please explain why **having another student validate your contribution to a discussion** makes your anxiety better.

During groupwork, how, if at all, **does forming new personal connections with other students** affect your feelings of anxiety?

It doesn't affect my anxiety	Makes my anxiety a little better	Makes my anxiety somewhat better	Makes my anxiety a lot better	I've never experienced this.
0	1	2	3	X



Students who answered 2 or greater were displayed this question:

Please explain why you feel that **forming new personal connections with other students** makes your anxiety better.

The next set of questions will ask you to what extent aspects of groupwork make your feelings of anxiety **worse**. When we refer to feelings of anxiety, we are talking about feeling nervous, anxious, on edge,

afraid, or worried. Please answer the following questions based on your experience in the one science course you indicated at the beginning of the survey.

During groupwork, how, if at all, **does being the center of attention** affect your feelings of anxiety?

It doesn't affect my anxiety	Makes my anxiety a little worse	Makes my anxiety somewhat worse	Makes my anxiety a lot worse	I've never experienced this.
------------------------------------	--	--	------------------------------------	------------------------------------

0	1	2	3	X
---	---	---	---	---



Students who answered 2 or greater were displayed this question:

Please explain why **being the center of attention** makes your anxiety worse.

During groupwork, how, if at all, **does realizing that you may struggle with content more than other students** affect your feelings of anxiety?

It doesn't affect my anxiety	Makes my anxiety a little worse	Makes my anxiety somewhat worse	Makes my anxiety a lot worse	I've never experienced this.
------------------------------------	--	--	------------------------------------	------------------------------------

0	1	2	3	X
---	---	---	---	---



Students who answered 2 or greater were displayed this question:

Please explain why **realizing that you may struggle with content more than other students** makes your anxiety worse.

During groupwork, how, if at all, **does having your grade depend on other students' contributions** affect your feelings of anxiety?

It doesn't affect my anxiety	Makes my anxiety a little worse	Makes my anxiety somewhat worse	Makes my anxiety a lot worse	I've never experienced this.
------------------------------------	--	--	------------------------------------	------------------------------------

0	1	2	3	X
---	---	---	---	---



Students who answered 2 or greater were displayed this question:

Please explain why **having your grade depend on other students' contributions** makes your anxiety worse.

During groupwork, how, if at all, **does working with other students who you do NOT know** affect your feelings of anxiety?

It doesn't affect my anxiety	Makes my anxiety a little worse	Makes my anxiety somewhat worse	Makes my anxiety a lot worse	I've never experienced this.
------------------------------	---------------------------------	---------------------------------	------------------------------	------------------------------

0	1	2	3	X
---	---	---	---	---



Students who answered 2 or greater were displayed this question:

Please explain why **working with other students who you do NOT know** makes your anxiety worse.

During groupwork, how, if at all, **does making a mistake in front of other students** affect your feelings of anxiety?

It doesn't affect my anxiety	Makes my anxiety a little worse	Makes my anxiety somewhat worse	Makes my anxiety a lot worse	I've never experienced this.
------------------------------	---------------------------------	---------------------------------	------------------------------	------------------------------

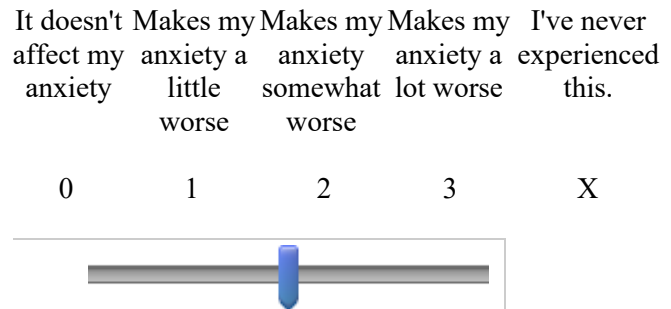
0	1	2	3	X
---	---	---	---	---



Students who answered 2 or greater were displayed this question:

Please explain why **making a mistake in front of other students** makes your anxiety worse.

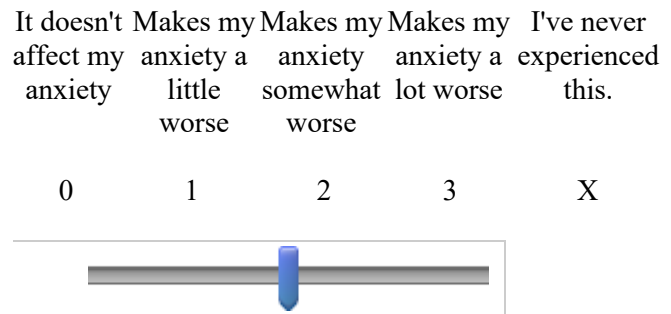
During groupwork, how, if at all, **does feeling like you have talked too much** affect your feelings of anxiety?



Students who answered 2 or greater were displayed this question:

Please explain why **feeling like you have talked too much** makes your anxiety worse.

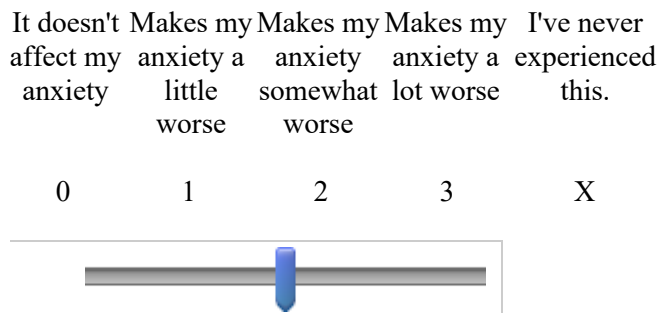
During groupwork, how, if at all, **does feeling like you have talked too LITTLE** affect your feelings of anxiety?



Students who answered 2 or greater were displayed this question:

Please explain why **feeling like you have talked too LITTLE** makes your anxiety worse.

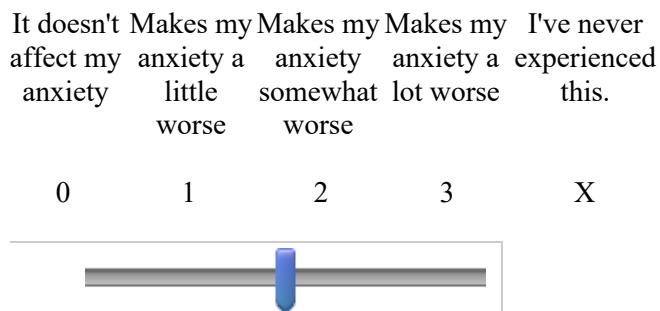
During groupwork, how, if at all, **does a student responding rudely to your contribution** affect your feelings of anxiety?



Students who answered 2 or greater were displayed this question:

Please explain why **a student responding rudely to your contribution** makes your anxiety worse.

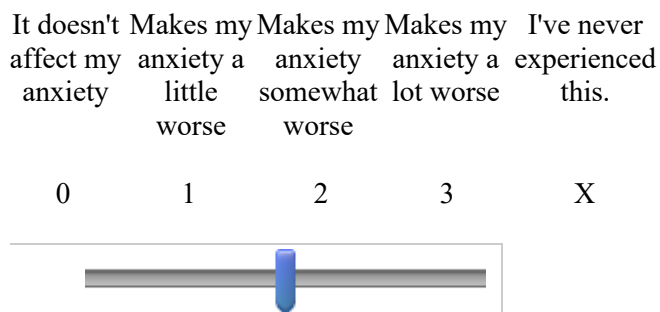
During groupwork, how, if at all, **does other students treating you differently than they would treat others** (potentially based on identity, abilities, appearance, personality, etc...) affect your feelings of anxiety?



Students who answered 2 or greater were displayed this question:

Please explain why **other students treating you differently than they would treat others** makes your anxiety worse.

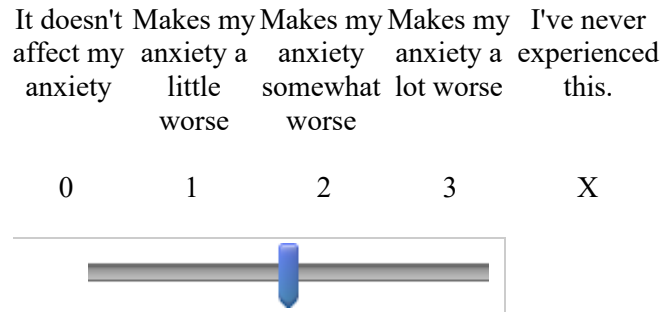
During groupwork, how, if at all, **does being asked to initiate social interactions (find a group)** affect your feelings of anxiety?



Students who answered 2 or greater were displayed this question:

Please explain why **being asked to initiate social interactions (find a group)** makes your anxiety worse.

During groupwork, how, if at all, **does not being able to be your authentic self** affect your feelings of anxiety?



Students who answered 2 or greater were displayed this question:

Please explain why **not being able to be your authentic self** makes your anxiety worse.

Please indicate how your **feelings of anxiety** impact **your ability to do the following during groupwork** in your college science course.

For example, for the first row you should indicate whether your anxiety makes it harder or easier for you to remember content that you learned during groupwork.

	Makes it a lot harder	Makes it a little harder	Doesn't change my ability	Makes it a little easier	Makes it a lot easier
Remember content that you learned	☐	☐	☐	☐	☐
Stay focused during groupwork	☐	☐	☐	☐	☐
Be motivated to engage in discussions during groupwork	☐	☐	☐	☐	☐
Be patient with others during groupwork	☐	☐	☐	☐	☐
Communicate your thoughts during groupwork	☐	☐	☐	☐	☐
Your willingness to contribute during groupwork	☐	☐	☐	☐	☐
Understand what others are saying during groupwork	☐	☐	☐	☐	☐
Solving problems during groupwork	☐	☐	☐	☐	☐
Feel confident about your intelligence during groupwork	☐	☐	☐	☐	☐

In your opinion, what can primary instructors do to help decrease students' feelings of anxiety associated with groupwork?

Our lab group focuses on creating a more inclusive scientific environment for all students. As such, it is helpful for us to know more about you. You may skip any question you do not want to answer.

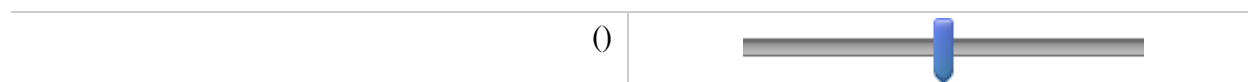
What field is your major most closely associated with? If you have more than one major, please choose all that apply.

- Biology
- Physics
- Geoscience
- Chemistry or biochemistry
- Engineering or technology
- Nursing, public health, or medical studies
- Business
- Literature, art, or design
- Another field not listed here _____

What is your GPA (on a 4.0 scale)? If this is your first semester, please select "no GPA".

no GPA

0 1 2 3 4



I most closely identify as

- Woman
- Man
- Gender-queer or non-binary
- An identity not listed, please specify _____
- Decline to state

I most closely identify as

- American Indian or Alaska Native
- Asian or Asian American
- Black or African American
- Middle Eastern or Middle Eastern American
- Hispanic, Latina/e, or of Spanish origin
- Pacific Islander
- White
- An identity not listed including multiracial, please describe

- Decline to state

What is your parent's highest completed level of education? If you have more than one parent with differing levels of education, choose the higher of the two.

- Did not complete high school
- High school diploma or GED
- Some college but no degree
- Associate degree (for example: AA, AS)
- Bachelor's degree (for example: BA, AB, BS)
- Master's degree (for example: MA, MS, MEng, MEd, MSW, MBA)
- Higher than a Master's degree (for example: PhD, MD, JD)
- Other (please describe)
○ _____
- Decline to state

Do you currently feel financially stable (e.g. have enough money to pay for necessities such as food and rent).

- Yes
- Yes, but only sometimes
- No
- Decline to state

Do you identify as a member of the LGBTQIA+* community?

**lesbian, gay, bisexual, transgender, queer/questioning, intersex, asexual, and more*

- Yes
- No
- Decline to state

I most closely identify with

- Currently or having previously struggled with anxiety or an anxiety disorder
- I have never struggled with an anxiety disorder
- Decline to state

RQ1a: output for final cross-validation models

Raw output of final models from cross-validation selection models for RQ1a outcomes; aligns with Table 1 of the main text.

A cross-validated selection procedure was run on each unique outcome, starting with identical predictors and model settings.

Initial foundational model:

```
initial.model.template <- caret::train( [outcome] ~ gender +  
lgbtq + race + firstgen + gpa.mean.centered + finance +  
class.size.scale + class.discipline, data = data.RQ1.analysis,  
method = "lmStepAIC", direction = "backward", trControl = ctrl)
```

Being able to work with other students who you know

Call:

```
lm(formula = .outcome ~ lgbtqLGBTQ + racePEER + gpa.mean.centered +  
class.disciplineChemistry, data = dat)
```

Residuals:

Min	1Q	Median	3Q	Max
-2.0111	-0.5300	0.2299	0.5273	1.5295

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	1.6816	0.1015	16.569	< 2e-16 ***
lgbtqLGBTQ	0.4049	0.1679	2.412	0.01717 *
racePEER	0.2940	0.1455	2.021	0.04524 *
gpa.mean.centered	0.5922	0.1695	3.494	0.00064 ***
class.disciplineChemistry	0.3615	0.1846	1.958	0.05227 .

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.8457 on 139 degrees of freedom

Multiple R-squared: 0.1626, Adjusted R-squared: 0.1385

F-statistic: 6.746 on 4 and 139 DF, p-value: 5.429e-05

Having another student validate your contribution to a discussion

Call:

```
lm(formula = .outcome ~ 1, data = dat)
```

Residuals:

Min	1Q	Median	3Q	Max
-1.95833	-0.75833	0.04167	1.04167	1.04167

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	1.95833	0.07512	26.07	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.9014 on 143 degrees of freedom

Realizing that other students struggle with the content too

Call:

```
lm(formula = .outcome ~ genderWoman.NB + lgbtqLGBTQ + raceAsian +  
    gpa.mean.centered, data = dat)
```

Residuals:

Min	1Q	Median	3Q	Max
-2.3578	-0.5427	0.1863	0.7752	1.3582

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	1.6973	0.1540	11.022	<2e-16	***
genderWoman.NB	0.3028	0.1724	1.757	0.0811	.
lgbtqLGBTQ	0.3340	0.1840	1.815	0.0716	.
raceAsian	-0.3619	0.1998	-1.812	0.0722	.
gpa.mean.centered	0.3958	0.1870	2.117	0.0360	*

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.9352 on 139 degrees of freedom

Multiple R-squared: 0.09906, Adjusted R-squared: 0.07313

F-statistic: 3.821 on 4 and 139 DF, p-value: 0.0056

Forming new personal connections with other students

Call:

```
lm(formula = .outcome ~ genderWoman.NB + racePEER + gpa.mean.centered,  
    data = dat)
```

Residuals:

Min	1Q	Median	3Q	Max
-1.9313	-0.5659	0.1432	0.5437	1.6347

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	1.4200	0.1397	10.161	< 2e-16 ***
genderWoman.NB	0.2681	0.1568	1.710	0.08942 .
racePEER	0.4149	0.1461	2.840	0.00518 **
gpa.mean.centered	0.3901	0.1708	2.284	0.02387 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.8489 on 140 degrees of freedom

Multiple R-squared: 0.1056, Adjusted R-squared: 0.08648

F-statistic: 5.512 on 3 and 140 DF, p-value: 0.001315

Having other students support you academically

Call:

```
lm(formula = .outcome ~ gpa.mean.centered, data = dat)
```

Residuals:

Min	1Q	Median	3Q	Max
-1.8041	-0.6577	0.1958	0.5389	1.4246

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	1.67267	0.07397	22.614	<2e-16 ***
gpa.mean.centered	0.28589	0.17679	1.617	0.108

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.8871 on 142 degrees of freedom

Multiple R-squared: 0.01808, Adjusted R-squared: 0.01117

F-statistic: 2.615 on 1 and 142 DF, p-value: 0.1081

RQ2a: output for final cross-validation models

Raw output of final models from cross-validation selection models for RQ2a outcomes; aligns with Table 3 of the main text.

A cross-validated selection procedure was run on each unique outcome, starting with identical predictors and model settings.

Initial foundational model:

```
initial.model.template <- caret::train( [outcome] ~ gender +  
lgbtq + race + firstgen + gpa.mean.centered + finance +  
class.size.scale + class.discipline, data = data.RQ2.analysis,  
method = "lmStepAIC", direction = "backward", trControl = ctrl)
```

Having your grade depend on other students' contributions

Call:

```
lm(formula = .outcome ~ firstgenYes + class.size.scale, data =  
data.RQ2.analysis)
```

Residuals:

Min	1Q	Median	3Q	Max
-2.04982	-0.62009	0.01733	0.79930	1.65144

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	1.97081	0.11123	17.718	<2e-16 ***
firstgenYes	-0.43131	0.17811	-2.422	0.0173 *
class.size.scale	-0.13523	0.08805	-1.536	0.1278

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.8738 on 99 degrees of freedom

Multiple R-squared: 0.0749, Adjusted R-squared: 0.05621

F-statistic: 4.008 on 2 and 99 DF, p-value: 0.0212

A student responding rudely to your contribution

Call:

```
lm(formula = .outcome ~ genderWoman.NB + lgbtqLGBTQ + racePEER +  
    firstgenYes + financeUnstable + class.disciplinePhysics +  
    class.disciplineChemistry, data = data.RQ2.analysis)
```

Residuals:

Min	1Q	Median	3Q	Max
-1.76005	-0.64571	0.00942	0.63876	1.96790

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	0.6958	0.2327	2.990	0.003557	**
genderWoman.NB	0.6268	0.1943	3.226	0.001727	**
lgbtqLGBTQ	0.5906	0.2077	2.844	0.005467	**
racePEER	0.2762	0.1868	1.478	0.142734	
firstgenYes	-0.7290	0.1998	-3.648	0.000434	***
financeUnstable	0.5197	0.1898	2.738	0.007387	**
class.disciplinePhysics	0.3363	0.2107	1.596	0.113803	
class.disciplineChemistry	0.3469	0.2371	1.463	0.146787	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.8437 on 94 degrees of freedom

Multiple R-squared: 0.2737, Adjusted R-squared: 0.2196

F-statistic: 5.06 on 7 and 94 DF, p-value: 6.676e-05

Realizing that you may struggle with content more than other students

Call:

```
lm(formula = .outcome ~ genderWoman.NB + class.size.scale, data =  
data.RQ2.analysis)
```

Residuals:

Min	1Q	Median	3Q	Max
-2.08391	-0.81533	0.04815	0.89717	2.04300

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	1.03629	0.17383	5.961	3.86e-08	***
genderWoman.NB	0.76742	0.20569	3.731	0.000318	***
class.size.scale	-0.19868	0.09624	-2.064	0.041598	*

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.9525 on 99 degrees of freedom

Multiple R-squared: 0.1471, Adjusted R-squared: 0.1298

F-statistic: 8.535 on 2 and 99 DF, p-value: 0.0003805

Other students treating you differently than they would treat others

Call:

```
lm(formula = .outcome ~ genderWoman.NB + lgbtqLGBTQ + raceAsian +  
    racePEER + financeUnstable + class.size.scale, data =  
data.RQ2.analysis)
```

Residuals:

Min	1Q	Median	3Q	Max
-1.82301	-0.48859	0.03966	0.46389	2.21346

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	0.48711	0.18966	2.568	0.01178	*
genderWoman.NB	0.64374	0.15556	4.138	7.57e-05	***
lgbtqLGBTQ	0.47879	0.17449	2.744	0.00726	**
raceAsian	0.41705	0.19418	2.148	0.03428	*
racePEER	0.26129	0.16431	1.590	0.11510	
financeUnstable	0.37823	0.15447	2.449	0.01618	*
class.size.scale	-0.21804	0.07436	-2.932	0.00422	**

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.7194 on 95 degrees of freedom

Multiple R-squared: 0.3361, Adjusted R-squared: 0.2942

F-statistic: 8.015 on 6 and 95 DF, p-value: 5.22e-07

Making a mistake in front of other students

Call:

```
lm(formula = .outcome ~ 1, data = data.RQ2.analysis)
```

Residuals:

Min	1Q	Median	3Q	Max
-1.48529	-0.58529	0.01471	0.51471	1.51471

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	1.48529	0.09011	16.48	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.9101 on 101 degrees of freedom

Being the center of attention

Call:

```
lm(formula = .outcome ~ genderWoman.NB, data = data.RQ2.analysis)
```

Residuals:

Min	1Q	Median	3Q	Max
-1.4718	-0.5718	-0.0714	0.6034	1.8290

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	1.1710	0.1668	7.022	2.69e-10 ***
genderWoman.NB	0.3009	0.1999	1.505	0.135

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.9285 on 100 degrees of freedom

Multiple R-squared: 0.02216, Adjusted R-squared: 0.01238

F-statistic: 2.266 on 1 and 100 DF, p-value: 0.1354

Being asked to initiate social interactions

Call:

```
lm(formula = .outcome ~ genderWoman.NB + financeUnstable, data =  
data.RQ2.analysis)
```

Residuals:

Min	1Q	Median	3Q	Max
-1.6685	-0.7087	-0.0953	0.6750	1.6663

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.7872	0.2051	3.838	0.000219 ***
genderWoman.NB	0.5465	0.1958	2.791	0.006314 **
financeUnstable	0.3349	0.1925	1.739	0.085112 .

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.9088 on 99 degrees of freedom

Multiple R-squared: 0.1021, Adjusted R-squared: 0.08401

F-statistic: 5.632 on 2 and 99 DF, p-value: 0.004827

Working with other students who you do not know

Call:

```
lm(formula = .outcome ~ raceAsian + racePEER + firstgenYes, data =  
data.RQ2.analysis)
```

Residuals:

Min	1Q	Median	3Q	Max
-1.46632	-0.50717	0.03148	0.54028	1.87921

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	1.1208	0.1383	8.104	1.52e-12 ***
raceAsian	0.3389	0.2167	1.564	0.1211
racePEER	0.3455	0.1997	1.730	0.0867 .
firstgenYes	-0.2956	0.1821	-1.623	0.1078

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.8147 on 98 degrees of freedom

Multiple R-squared: 0.04984, Adjusted R-squared: 0.02075

F-statistic: 1.713 on 3 and 98 DF, p-value: 0.1692

Feeling like you have talked too much

Call:

```
lm(formula = .outcome ~ lgbtqLGBTQ + raceAsian + gpa.mean.centered,  
    data = data.RQ2.analysis)
```

Residuals:

Min	1Q	Median	3Q	Max
-1.55463	-0.76551	0.03375	0.54827	2.06243

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.8974	0.1059	8.473	2.45e-13 ***
lgbtqLGBTQ	0.4737	0.2044	2.317	0.0226 *
raceAsian	0.3510	0.2031	1.728	0.0872 .
gpa.mean.centered	-0.2868	0.2049	-1.399	0.1649

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.8491 on 98 degrees of freedom

Multiple R-squared: 0.09126, Adjusted R-squared: 0.06344

F-statistic: 3.281 on 3 and 98 DF, p-value: 0.02414

Not being able to be your authentic self

Call:

```
lm(formula = .outcome ~ class.size.scale, data = data.RQ2.analysis)
```

Residuals:

Min	1Q	Median	3Q	Max
-1.25687	-0.64660	-0.04993	0.54333	2.08075

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	1.03319	0.08387	12.319	<2e-16 ***
class.size.scale	-0.15535	0.08356	-1.859	0.0659 .

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.8295 on 100 degrees of freedom

Multiple R-squared: 0.03341, Adjusted R-squared: 0.02375

F-statistic: 3.457 on 1 and 100 DF, p-value: 0.06593

Feeling like you have talked too little

Call:

```
lm(formula = .outcome ~ genderWoman.NB + financeUnstable, data =  
data.RQ2.analysis)
```

Residuals:

Min	1Q	Median	3Q	Max
-1.16303	-0.55242	-0.00068	0.54181	2.17675

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.4808	0.1643	2.926	0.00426 **
genderWoman.NB	0.3425	0.1569	2.183	0.03141 *
financeUnstable	0.3398	0.1543	2.203	0.02993 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.7281 on 99 degrees of freedom

Multiple R-squared: 0.09226, Adjusted R-squared: 0.07392

F-statistic: 5.031 on 2 and 99 DF, p-value: 0.0083

RQ3a: output for final stepwise selection models

Raw output of final models for stepwise ordinal regressions for RQ3 outcomes; aligns with Table 5 in the main text.

```
ordmod.outcome <- ordinal::clm(outcome.fact ~ gender + lgbtq +  
  race + firstgen + gpa.mean.centered + finance + class.size.scale  
  + class.discipline + gad.avg, data = data.RQ3.analysis, Hess =  
  TRUE)
```

```
step.ord.outcome <- step(ordmod.outcome, direction = "backward")
```

Communicate your thoughts

```
formula: AonAL_communicate.fact ~ gender + firstgen + class.size.scale  
+ class.discipline  
data:    data.RQ3.analysis
```

```
link threshold nobs logLik AIC      niter max.grad cond.H  
logit flexible  178  -249.21 516.43 5(0)   8.55e-08 3.5e+01
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
genderWoman.NB	-0.4227	0.2902	-1.457	0.1452
firstgenYes	0.5396	0.2949	1.830	0.0673 .
class.size.scale	0.4171	0.1622	2.571	0.0101 *
class.disciplinePhysics	0.8763	0.3886	2.255	0.0241 *
class.disciplineChemistry	0.4175	0.4436	0.941	0.3467

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Threshold coefficients:

	Estimate	Std. Error	z value
-2 -1	-2.08642	0.33160	-6.292
-1 0	0.07327	0.27285	0.269
0 1	1.42321	0.29559	4.815
1 2	2.82796	0.38601	7.326

Feel confident about your intelligence

formula:

```
AonAL_confident.fact ~ gender + firstgen + gpa.mean.centered +  
class.size.scale + class.discipline
```

data: data.RQ3.analysis

```
link threshold nobs logLik AIC niter max.grad cond.H  
logit flexible 178 -262.04 544.08 5(0) 6.29e-13 3.6e+01
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
genderWoman.NB	-0.8287	0.2908	-2.850	0.00438	**
firstgenYes	0.7673	0.2972	2.582	0.00983	**
gpa.mean.centered	0.5810	0.3293	1.764	0.07769	.
class.size.scale	0.3258	0.1609	2.025	0.04287	*
class.disciplinePhysics	0.7871	0.3929	2.003	0.04515	*
class.disciplineChemistry	0.1734	0.4412	0.393	0.69440	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Threshold coefficients:

	Estimate	Std. Error	z value
-2 -1	-1.6103	0.2993	-5.381
-1 0	-0.1184	0.2662	-0.445
0 1	1.2096	0.2799	4.321
1 2	2.0421	0.3218	6.346

Be willing to contribute

formula: AonAL_contribute.fact ~ firstgen

data: data.RQ3.analysis

link	threshold	nobs	logLik	AIC	niter	max.grad	cond.H
logit	flexible	178	-265.99	541.99	5(0)	2.18e-11	1.5e+01

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
firstgenYes	0.5397	0.2893	1.866	0.0621

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Threshold coefficients:

	Estimate	Std. Error	z value
-2 -1	-2.1593	0.2740	-7.882
-1 0	-0.5373	0.1815	-2.961
0 1	0.9647	0.1920	5.025
1 2	2.1118	0.2519	8.384

Be motivated to engage in discussions

formula: AonAL_engage.fact ~ class.size.scale + gad.avg

data: data.RQ3.analysis

```
link threshold nobs logLik AIC niter max.grad cond.H
logit flexible 178 -263.67 539.34 5(0) 2.12e-11 1.8e+01
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
class.size.scale	0.2796	0.1418	1.972	0.0487 *
gad.avg	-0.2886	0.1791	-1.612	0.1070

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Threshold coefficients:

	Estimate	Std. Error	z value
-2 -1	-1.9808	0.2614	-7.578
-1 0	-0.2346	0.2043	-1.148
0 1	0.9723	0.2233	4.354
1 2	2.1622	0.3017	7.166

Stay focused

```
formula: AonAL_focus.fact ~ gender + race + firstgen +  
class.size.scale + class.discipline + gad.avg  
data:    data.RQ3.analysis
```

```
link threshold nobs logLik AIC      niter max.grad cond.H  
logit flexible 178  -242.05 508.10 5(0)  5.54e-07 7.1e+01
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
genderWoman.NB	-0.7899	0.3008	-2.625	0.008652	**
raceAsian	1.0633	0.3797	2.800	0.005111	**
racePEER	0.8758	0.3352	2.613	0.008988	**
firstgenYes	0.4877	0.3116	1.565	0.117515	
class.size.scale	0.4773	0.1741	2.742	0.006108	**
class.disciplinePhysics	0.7691	0.4048	1.900	0.057411	.
class.disciplineChemistry	1.6933	0.4534	3.735	0.000188	***
gad.avg	-0.2750	0.1819	-1.512	0.130613	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Threshold coefficients:

	Estimate	Std. Error	z value
-2 -1	-2.1799	0.3938	-5.535
-1 0	0.1067	0.3388	0.315
0 1	1.6424	0.3608	4.553
1 2	2.9402	0.4268	6.889

Be patient with others

```
formula: AonAL_patient.fact ~ race + firstgen + gpa.mean.centered +  
gad.avg
```

```
data: data.RQ3.analysis
```

```
link threshold nobis logLik AIC niter max.grad cond.H  
logit flexible 178 -245.20 508.41 5(0) 1.03e-07 3.6e+01
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
raceAsian	0.7922	0.3775	2.098	0.0359 *
racePEER	-0.3109	0.3203	-0.971	0.3317
firstgenYes	0.6528	0.3081	2.119	0.0341 *
gpa.mean.centered	-0.6471	0.3395	-1.906	0.0567 .
gad.avg	-0.3274	0.1738	-1.884	0.0595 .

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Threshold coefficients:

	Estimate	Std. Error	z value
-2 -1	-2.4504	0.3457	-7.088
-1 0	-0.5672	0.2662	-2.131
0 1	1.3181	0.2857	4.614
1 2	2.4767	0.3529	7.018

Remember content that you learned

```
formula: AonAL_remember.fact ~ race + gad.avg
```

```
data:    data.RQ3.analysis
```

```
link threshold nobs logLik AIC      niter max.grad cond.H
logit flexible 178  -258.81 531.61 5(0)  1.87e-10 3.3e+01
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
raceAsian	0.6567	0.3684	1.783	0.0747	.
racePEER	-0.1675	0.3037	-0.552	0.5813	
gad.avg	-0.3673	0.1827	-2.011	0.0443	*

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Threshold coefficients:

	Estimate	Std. Error	z value
-2 -1	-2.0899	0.3144	-6.647
-1 0	-0.2725	0.2532	-1.076
0 1	1.0631	0.2707	3.927
1 2	2.3800	0.3561	6.683

Solve problems

```
formula: AonAL_solve.fact ~ gender + firstgen + class.size.scale  
data:    data.RQ3.analysis
```

```
link threshold nobs logLik AIC      niter max.grad cond.H  
logit flexible 178  -249.09 512.19 6(0)  1.23e-13 2.9e+01
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
genderWoman.NB	-0.7694	0.2902	-2.652	0.00801	**
firstgenYes	0.6001	0.2972	2.020	0.04343	*
class.size.scale	0.2234	0.1461	1.529	0.12631	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Threshold coefficients:

	Estimate	Std. Error	z value
-2 -1	-3.2551	0.3973	-8.193
-1 0	-1.0130	0.2618	-3.869
0 1	0.6989	0.2526	2.767
1 2	1.5218	0.2840	5.359

Understand what others are saying

```
formula: AonAL_understand.fact ~ gender + race + firstgen +  
class.size.scale  
data:    data.RQ3.analysis
```

```
link threshold nobs logLik AIC      niter max.grad cond.H  
logit flexible 178  -244.89 507.78 6(0)   5.05e-11 3.3e+01
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
genderWoman.NB	-0.6136	0.2881	-2.130	0.0332 *
raceAsian	0.7287	0.3740	1.949	0.0513 .
racePEER	0.5187	0.3269	1.587	0.1126
firstgenYes	0.4350	0.3076	1.414	0.1574
class.size.scale	0.2341	0.1487	1.574	0.1154

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Threshold coefficients:

	Estimate	Std. Error	z value
-2 -1	-3.0846	0.4367	-7.063
-1 0	-0.5866	0.2895	-2.026
0 1	1.0517	0.2982	3.527
1 2	2.3377	0.3566	6.556

Justification for variable grouping

We combined students who identified as nonbinary (NB) with women into one category to avoid excluding the small number of nonbinary individual; compared to men, both women and NB individuals are underserved in science (Cooper et al., 2020). Similarly, we grouped American Native or Pacific Islander, Black, and Multiracial students with Hispanic students in the persons excluded because of their ethnicity or race (PEER) category (Asai, 2020), and categorized Asian and white students separately.

Demographics table

Demographic		% (n) (N= 272)
Gender	Man	33% (89)
	Woman	63% (172)
	Non-binary or gender queer	3% (8)
	<i>Decline to state</i>	1% (4)
Race/ethnicity	American Native or Pacific Islander	2% (5)
	Asian	19% (53)
	Black	1% (3)
	Hispanic	26% (71)
	Multiracial	4% (10)
	White	40% (110)
	<i>Decline to state</i>	5% (14)
LGBTQ+	No	75% (203)
	Yes	21% (56)
	<i>Decline to state</i>	5% (13)
College Generation Status	Continuing	61% (165)
	First-generation	35% (96)
	<i>Decline to state</i>	4% (11)
Financial Stable	Yes	43% (116)
	No	52% (141)
	<i>Decline to state</i>	5% (15)
GPA	Mean	3.54
	"+- std"	0.45
Major	STEM major	69% (188)
	Non-STEM	29% (80)
	<i>Decline to state</i>	1% (4)
	Mean (range: 0-3)	0.82
Anxiety (GAD-7, mean scores) ^	"+- std"	0.8
	Mean (range: 0-3)	0.82
Trait anxiety severity (GAD-7, sum scores) ^^	None or minimal	52% (141)
	Mild	25% (68)
	Moderate	13% (35)
	Severe	10% (26)
	<i>Decline to state</i>	1% (2)
	Biology	68% (186)
Class discipline	Chemistry	11% (31)
	Physics	14% (39)
	<i>Decline to state</i>	6% (16)
	Mean	159.42
Class size	"+- std"	101.69

^Mean scores were used in all regression analyses as mean scores are more robust to low levels of missing data (e.g., a participant accidentally skips one item in a 7-item scale) than sum scores (Parent, 2013).

^^ Based on clinical cutoffs of GAD-7 sum scores (Kroenke et al., 2007; Plummer et al., 2016; Spitzer et al., 2006).

Additional codes for why aspects of groupwork alleviate students' anxiety

Themes	Description	Example quote
Provides joy	Student describes feeling happier and joyful.	Student 184 (Forming new personal connections): "Making new connections with other students is fun and enjoyable."
Boosts confidence	Student describes feeling more confident.	Student 82 (Receiving validation): "It makes me feel more confident in my abilities in the class."
Cultivates trust	Student describes feelings of trust and reliability.	Student 229 (Working with student you know): "I much prefer working with others I know as there a sense of camaraderie."
Maintains accountability	Student describes feeling held accountable to stay on track and succeed.	Student 27 (Receiving student academic support): "I worry much less about missing important deadlines or assignments if I have other students that can support me in the class."

Additional codes for why aspect of groupwork exacerbate students' anxiety

Themes	Description	Example quote
Dread exclusion	Student describes worrying about being rejected, left out and excluded.	Student 34 (Being asked to initiate social interactions): "I don't want to run the risk of being rejected by a group."
Engenders hopelessness	Student describes feelings of hopelessness, lack of belonging and uncertainty about the likelihood of future success.	Student 257 (Realizing you struggle more than others): "It is validating my feelings that I am not worthy to be in college and am going to fail."
Generates external pressure	Student describes feeling an external expectation and pressures from others and the need to meet them.	Student 268 (Having grade dependent on others' contributions): "What if my part is bad or I didn't do it right then that means [my peers] grade is on me as well and if they are doing really well and their grade drops it's because of me and it is my fault."

Additional student recommendations for how instructors can decrease anxiety

Themes	Description	Percent (n=249)	Example quote
Initiate conversations	Students recommend breaking the ice, initiating, encouraging and facilitating conversations between group members.	8.0% (20)	Student 18: "Encourage friendly discussions among students."
Maintain accountability	Students recommend monitoring groups to maintain accountability, address conflicts or issues and ensure comfort and equal contributions.	6.8% (17)	Student 40: "Walk around and make sure that everyone is participating and not letting one person talk and everyone else doesn't engage."
Instruct better	Students recommend practicing, learning, teaching and understanding the content more or better and providing clearer instructions.	6.4% (16)	Student 98: "Explaining material better during lectures."
Provide resources	Students recommend providing support beyond group work, such as holding exam reviews, giving feedback on assignments, more clarity in syllabus, post reminders, offer extra credit, flexibility and extending deadlines.	5.6% (14)	Student 21: "Make everything extremely clear in the syllabus."
Provide support	Students recommend providing more support and guidance by checking in, being more involved, interactive and available to answer questions or help.	4.8% (12)	Student 117: "Be available to answer any questions."
Be personable	Students recommend being more open, kind, patient, reassuring, forgiving, authentic and not judgmental.	4.8% (12)	Student 220: "Bring understanding and forgiving. Knowing that I can go to them about problems and that they will be addressed!"
Welcome feedback	Students recommend giving them opportunities share anonymous feedback and opinions about the course, group, etc. through surveys.	3.6% (9)	Student 15: "Initiate a quiz like this to the students beforehand as an anonymous poll to see if group work would be more beneficial to the class than harmful to the student's anxiety."

Don't assign or reduce groupwork	Students recommend not assigning group work or reducing it.	2.4% (6)	Student 233: "Not assigning group work if possible."
Maintain consistency among groups	Students recommend keeping groups consistent.	2% (5)	Student 229: "Keep those groups throughout a semester."

Supplemental references

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